

Poker Equity Calculation App

Context

Poker is a game of incomplete information, but it is also a game of mathematics. Every decision at the table has an optimal answer rooted in probability. Most players rely on intuition built from years of experience, but the underlying math is computable. Live Equity is an iOS application that calculates a player's equity and hand strength in real time as cards are dealt, giving any player instant access to the same probabilistic reasoning that professionals have internalized over thousands of hours.

Data Structures and Algorithms

The two core components are a hash map for $O(1)$ hand evaluation and a Monte Carlo simulation for equity estimation.

Every card is encoded as an integer from 0-51 using $\text{rank} \times 4 + \text{suit}$. This reduces the entire deck to a simple integer array and makes card comparisons arithmetic. To evaluate a hand, all possible 5-card subsets are examined (up to $C(7,5) = 21$ with a full board), and each is scored by encoding its category and tiebreaker ranks into a single integer using bit-shifting. Any two hands can then be compared with one integer comparison.

Equity cannot be calculated exactly because opponents' cards are unknown. Instead, the app runs a Monte Carlo simulation: it deals thousands of random runouts, evaluates all hands, and tracks how often the player wins or ties. With 1,000 simulations, results are accurate to within roughly 2-3% and return in well under a second.

Why It's Cool

The app updates live with every card entered. Add the flop, turn, or river and equity recalculates instantly. For any poker player, this is a legitimate study tool where you can replay hands from a session and immediately see whether a fold or call was mathematically correct.

Technical Challenges

The trickiest challenge was the hand evaluator's integer encoding. Naively packing category and rank values into a single integer causes overlap between categories, where a high-card hand can produce a larger number than a low one-pair hand if the scaling is wrong. The solution was bit-shifting: the category occupies bits 20-23, and up to five rank nibbles (4 bits each) are packed into the lower 20 bits, guaranteeing strict ordering across all categories with no collisions.

The second challenge was UI responsiveness. Running 1,000 Monte Carlo simulations synchronously blocked the main thread, making card input feel sluggish. This was solved using Swift's `Task.detached` to run simulations on a background thread, with `MainActor.run` to safely push results back to the UI when complete.

Work Distribution

Caitlin primarily worked on the UI design and interface and Zack primarily worked on the algorithms. However before we got to the actual code, we spent a lot of time collectively working on designing the algorithms and figuring out what problems we would face, and how to effectively make an interface for fast and easy card input.

Github Repo:

<https://github.com/Zack-Weinstein/CF2PokerEquity.git>

Caitlin Owen
Zachary Weinstein
April 29th, 2026
Atlas 2270 "CF2"

15:41

Poker Equity

Players

2

3

4

5

6

7

8

9

Buy In

Small Blind

Big Blind

.25

.50

Start Game

15:42

40%

Live Equity

New Game

Players: 6

-

Your Equity

0.0%

Hand Strength

0.0%

Cards

-

-

-

-

-

-

-

-

Select Rank

2

3

4

5

6

7

8

9

10

J

Q

K

A

Select Suit

♠

♥

♦

♣

Clear Selected

Next Hand

15:42

63%

Live Equity

New Game

Players: 6

-

Your Equity

75.6%

Hand Strength

95.7%

Cards

5

♥

6

♦

7

♦

8

♦

9

♦

10

♦

-

Select Rank

2

3

4

5

6

7

8

9

10

J

Q

K

A

Select Suit

♠

♥

♦

♣

Clear Selected

Next Hand

15:43

Live Equity

New Game

Players: 6

-

Your Equity

77.9%

Hand Strength

95.6%

Cards

5

♥

6

♦

7

♦

8

♦

9

♦

10

♦

Q

♦

Select Rank

2

3

4

5

6

7

8

9

10

J

Q

K

A

Select Suit

♠

♥

♦

♣

Clear Selected

Next Hand

15:43

Live Equity

New Game

Players: 5

-

Your Equity

35.9%

Hand Strength

92.0%

Cards

K

♦

A

♥

A

♠

J

♠

Q

♣

-

-

Select Rank

2

3

4

5

6

7

8

9

10

J

Q

K

A

Select Suit

♠

♥

♦

♣

Clear Selected

Next Hand